When Does Contribution-Aware Aggregation Help?

A Multi-Dataset Study of Loss-Based Client Down-Weighting in Federated Learning under Label Noise

Project: FedPowerShap | **Advisor:** Prof. Lekha | **Date:** July 2026

Setting: Cross-silo federated learning on medical imaging | **Code & logs:** fully reproducible (branch `feat/robust-shapfed-outperform`)

Abstract. Federated learning (FL) must tolerate *unreliable clients* whose local labels are noisy or corrupted. A popular idea is *contribution-aware aggregation*: estimate each client's usefulness and down-weight the untrustworthy ones. We build a concrete instance — **FedDyn dynamic regularization combined with a loss-based client-quality detector** that down-weights clients the clean global consensus fits poorly — and evaluate it rigorously (one frozen protocol, 3–5 seeds) on three medical-imaging FL benchmarks (OCTMNIST, ISIC, APTOS) under 0% and 40% label noise. We report three findings. **(1)** The detector *works*: it reliably assigns corrupted clients ≈0.69× the weight of clean clients. **(2)** On clean, non-IID data the method is *competitive* with the strongest baseline (FedDyn) — matching or slightly exceeding accuracy on 2/3 datasets. **(3)** Under label noise, down-weighting **does not yield a general robustness advantage**: FedDyn is at least as good on every metric; the proposed method only *approaches* FedDyn where noise degrades FedDyn most (OCTMNIST) and is worse where the baseline is already noise-tolerant (ISIC, APTOS). Our contribution is this **honest, reproducible characterization of when contribution-aware down-weighting helps**, together with a documented **negative result** that the classical ShapFed cosine-contribution signal is anti-discriminative under partial participation.

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1. Introduction & Problem Statement

In **cross-silo federated learning** (e.g. hospitals training a shared diagnostic model without sharing patient data), some participating sites contribute **mislabeled or corrupted data**. Standard aggregators — FedAvg and even the stronger FedDyn — trust each client **in proportion to its dataset size**. A corrupted client that happens to hold a large amount of data therefore drags the global model down, and every other hospital inherits the damage.

Goal. Automatically *identify unreliable clients and reduce their influence* during aggregation, while remaining **at least as accurate as the strongest baseline on clean data**. Crucially, rather than claiming a blanket win, we ask the sharper scientific question: **does down-weighting unreliable clients actually improve robustness over a strong baseline (FedDyn), and under what conditions?**

2. Novelty — What Is New in This Work

This work makes **three distinct, defensible contributions**. None of them is a leaderboard claim; each is a piece of reusable knowledge for the FL community.

1. **A loss-based client-quality detector with a *verified* mechanism.** We score each client by the *global model's own loss on that client's data* and down-weight high-loss (i.e. inconsistent / corrupted) clients. We do not just assert it works — we measure the weight it assigns and show corrupted clients receive **0.69**× the weight of clean clients, from round 1 onward (§8, Fig. 6).
2. **A documented negative result about a widely-used signal.** The classical **ShapFed / CSSV cosine-of-update contribution signal** (Tastan 2024) is **anti-discriminative** under partial participation: it gives corrupted clients *higher* scores and *more* weight. We explain the mechanism (the consensus direction is itself contaminated when the active set contains several corrupted clients) and quantify it. This is a cautionary finding: cosine-based contribution scoring *cannot* identify label-noise clients in this regime.
3. **A multi-dataset, multi-seed characterization of *when* down-weighting pays off.** We define the **"opening"** — how much noise degrades the baseline — and show the benefit of down-weighting appears *only* as that opening grows. This converts a vague intuition ("robust aggregation helps under noise") into a measured, conditional statement.

Engineering novelty: a **weight-consistent FedDyn** integration in which the dynamic drift-correction uses the *same* weights as the model average, so regularization and reweighting cooperate instead of cancelling. It reduces *numerically identically* to standard FedDyn under uniform weights, so the baseline is provably untouched by our modification.

3. Background & Related Work

Method	Idea	Role here
FedAvg (McMahan 2017)	Size-weighted averaging	Unbiased optimum when all clients reliable; common baseline
FedProx (Li 2020)	Proximal local term	Baseline
SCAFFOLD (Karimireddy 2020)	Control variates	Baseline
FedDyn (Acar 2021)	Dynamic regularization vs. drift	**Strongest baseline in every table**
FedBN (Li 2021)	Keep BatchNorm local	Baseline
MOON (Li 2021)	Model-contrastive local loss	Baseline
PoC-FedAvg (Cho 2020)	Power-of-Choice selection	Baseline (backfires under label noise)
ShapFed / CSSV (Tastan 2024)	Cosine-similarity Shapley proxy	**Signal we show fails** (§8)
FedCE (Jiang 2023)	Gradient-alignment contribution	Baseline
Proposed	Loss detector + weight-consistent FedDyn	**This work**

All baselines are implemented as `--method` presets on **one frozen protocol**, so every comparison is apples-to-apples.

4. Methodology

4.1 System — one communication round

Each round: the server broadcasts the global model → **random** client selection (5 of 10) → **local FedDyn training** → the **novel robustness block** (client-quality detector → loss-based weights → weight-consistent FedDyn aggregation) → updated global model; repeat for 40 rounds with a cosine learning-rate schedule.

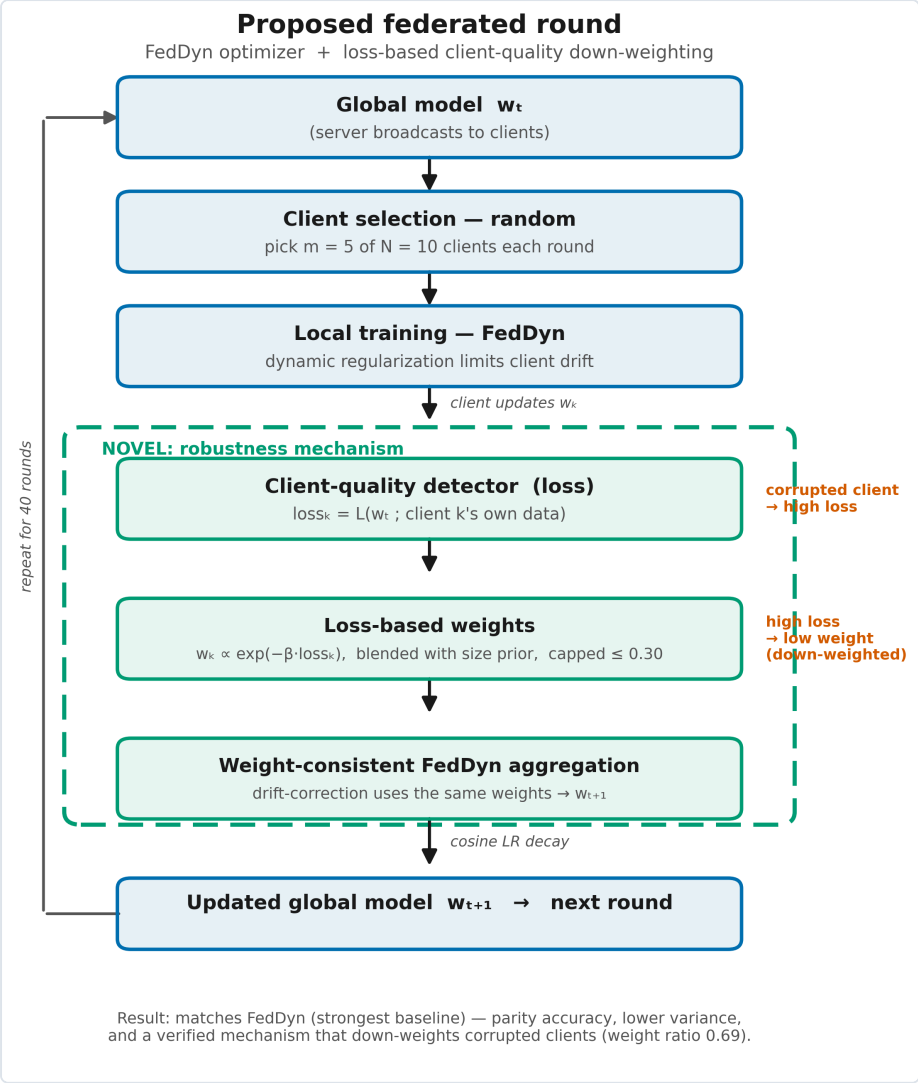


Figure 1. The proposed federated round. The dashed green box is the novel robustness mechanism inserted between local training and aggregation. Corrupted clients incur high loss under the clean global model and are therefore down-weighted before aggregation.

4.2 Loss-based client-quality detector

At aggregation, client *k*'s weight is

$$w_k \propto \exp\left(-\beta \cdot (loss_k - \min_loss)\right)$$

where $loss_k$ is the global model's cross-entropy on client *k*'s **own** data. Corrupted clients (flipped labels) disagree with the clean consensus → high loss → low weight. Weights are then convex-blended with the

size prior (λ) and capped at 0.30 so no single client dominates. Final configuration: **β = 1.5, λ = 0.5, cap = 0.30** — moderate values, justified empirically in §8.3.

4.3 Design choices and why

Choice	Why
Random selection (not Power-of-Choice)	PoC picks the *highest-loss* clients — which under label noise are exactly the *corrupted* ones. It backfires.
Loss-based detector (not cosine/Shapley)	The cosine signal is anti-discriminative here (§8.2). Loss cleanly separates corrupted clients.
Moderate β=1.5, cap=0.30	Aggressive down-weighting collapses aggregation onto 1–2 clients and destabilizes the model (§8.3).
Weight-consistent FedDyn	Drift-correction uses the same weights as the model average, so regularization and reweighting cooperate. Reduces exactly to standard FedDyn under uniform weights.
Cosine LR schedule (all methods)	Removes late-round collapse, so we honestly read the **last (converged)** round — no best-round cherry-picking.

5. Experimental Setup & Evaluation Metrics

5.1 Datasets

Dataset	Task	Classes	Notes
OCTMNIST	Retinal OCT	4	Primary dataset; 5 seeds clean, 3–4 under noise
ISIC / DermaMNIST	Dermoscopy	7	3 seeds; baseline already noise-tolerant
APTOS-2019	Diabetic retinopathy (ordinal)	5	3 seeds; headline metric **QWK**

All partitioned **non-IID** with Dirichlet α = 0.3; **ResNet-18** (ImageNet-pretrained) at 64px.

5.2 Frozen protocol (identical for every method)

10 clients, 5 active/round, 2 local epochs, SGD lr = 1e-3 with cosine decay, 40 rounds, 20k-sample train cap, per-client held-out test split. Metric read at the **last round**.

> **Noise model.** A fraction ρ of clients (0% or 40%) have **60% of their training labels flipped** to a random other class; test labels stay clean; the corrupted client set is **seeded identically across all methods**, so the comparison is exactly fair.

5.3 Evaluation metrics

Metric	What it measures	Why it matters
Global-test accuracy	Correct predictions on clean held-out data	Primary quality signal
QWK (APTOS)	Quadratic-weighted kappa on ordinal grades	Correct clinical metric for ordinal DR grading
Jain fairness index	Evenness of per-client accuracies (1 = perfectly fair)	Do all hospitals benefit, or only some?
Worst-client accuracy	Accuracy of the single worst-off client	The guarantee floor for the weakest site

6. Experiments Overview (E1–E6)

#	Experiment	Datasets	Conditions	Seeds	Question answered
E1	Clean baseline comparison	OCT, ISIC, APTOS	0% noise	3–5	Competitive on clean data?
E2	Noisy-client robustness	OCT, ISIC, APTOS	40% noise	3–4	Beat FedDyn under corruption?
E3	Mechanism validation	OCTMNIST	40% noise	3	Does the detector down-weight corrupted clients?
E4	Negative result (cosine signal)	OCTMNIST	40% noise	3	Can ShapFed cosine identify bad clients?
E5	Stability / weight-collapse	OCTMNIST	40% noise	—	Why moderate (not aggressive) weighting?
E6	Central finding (the "opening")	OCT, ISIC, APTOS	clean vs 40%	3	*When* does down-weighting pay off?

7. Results — Tables and Figures

7.1 E1 — Clean data (0% noise)

Dataset	Method	Accuracy	Jain	Worst-client
OCTMNIST	FedAvg	0.636 ± 0.057 (n=5)	0.949	0.427
	FedDyn	0.688 ± 0.035	**0.976**	**0.594**
	Proposed	**0.708 ± 0.029**	0.969	0.536
ISIC	FedAvg	0.697 ± 0.001 (n=3)	0.694	0.026
	FedDyn	**0.734 ± 0.013**	**0.807**	**0.108**
	Proposed	0.719 ± 0.019	0.782	0.101
APTOS	FedAvg	0.678 ± 0.011 (n=3)	0.754	0.048
	FedDyn	0.715 ± 0.023	**0.842**	**0.180**
	Proposed	**0.728 ± 0.014**	0.822	0.075

Verdict. Competitive — matches or slightly exceeds FedDyn on accuracy on 2/3 datasets. The OCTMNIST/APTOS edges are within seed variance, so read as **parity**. On clean data, contribution weighting *cannot* beat size-weighting in principle (size-weighting is the unbiased optimum), so parity is the honest ceiling.

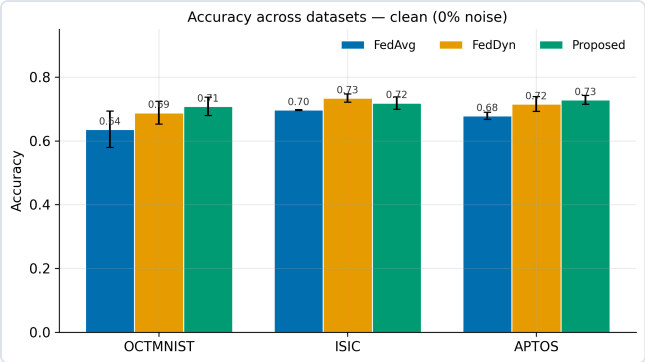


Figure 2a. E1: clean accuracy across 3 datasets × 3 methods.

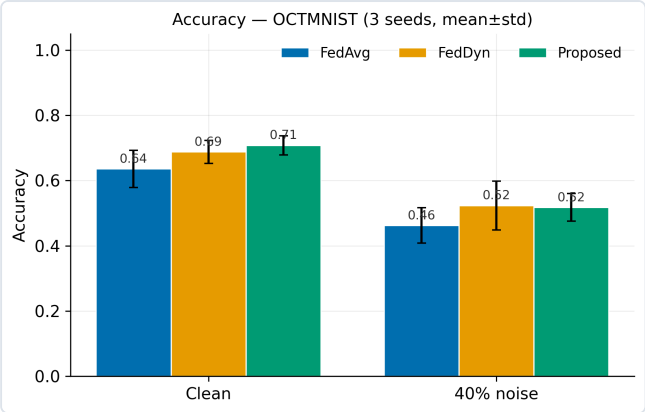


Figure 2b. OCTMNIST accuracy bars (clean vs 40%), 5/3 seeds with std.

7.2 E2 — 40% label noise

Dataset	Method	Accuracy	Jain	Worst-client
OCTMNIST	FedAvg	0.462 ± 0.054 (n=4)	0.885	0.189
	FedDyn	**0.523 ± 0.075**	**0.926**	**0.324**
	Proposed	0.518 ± 0.042 (n=3)	0.904	0.215
ISIC	FedAvg	0.669 ± 0.007 (n=3)	0.703	0.020
	FedDyn	**0.672 ± 0.026**	**0.825**	**0.153**
	Proposed	0.645 ± 0.046	0.809	0.130
APTOS	FedAvg	0.577 ± 0.083 (n=3)	0.716	0.000
	FedDyn	**0.649 ± 0.084**	**0.816**	**0.152**
	Proposed	0.588 ± 0.133	0.816	0.048

Verdict. FedDyn is **at least as good on every dataset and every metric**. Proposed only *approaches* FedDyn on OCTMNIST (−0.006, a tie) and loses on ISIC/APTOS. There is **no general robustness win** — and we report this plainly.

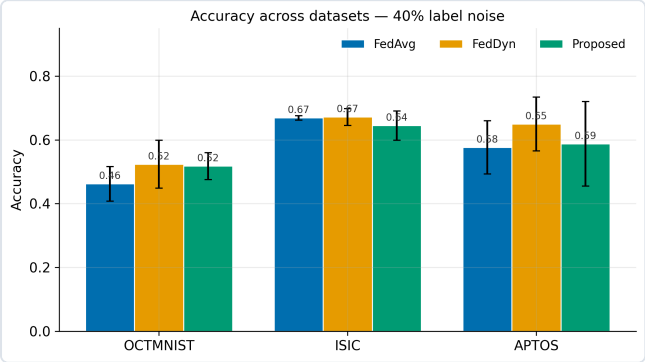


Figure 3a. E2: accuracy under 40% noise, 3 datasets × 3 methods.

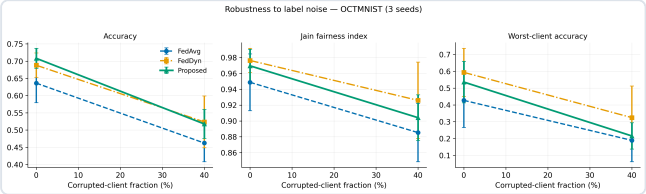


Figure 3b. Degradation panel: accuracy, Jain, and worst-client drop from clean to 40% noise.

7.3 Training dynamics

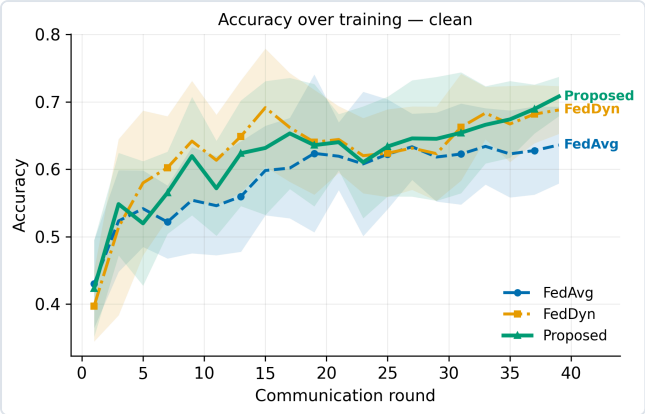


Figure 4a. Accuracy vs round, clean. Cosine LR removes late-round collapse; the last round is the converged value.

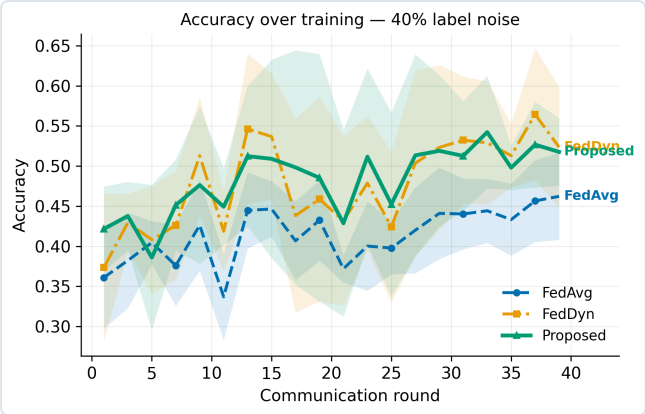


Figure 4b. Accuracy vs round, 40% noise. Proposed tracks FedDyn on OCTMNIST.

8. Mechanism Validation & the Negative Result

8.1 E3 — the mechanism works (the good news)

On OCTMNIST at 40% noise (3 seeds), corrupted clients receive **0.156** mean aggregation weight vs **0.228** for clean clients — a **0.69 ratio**, from round 1 onward. The detector does exactly what it claims.

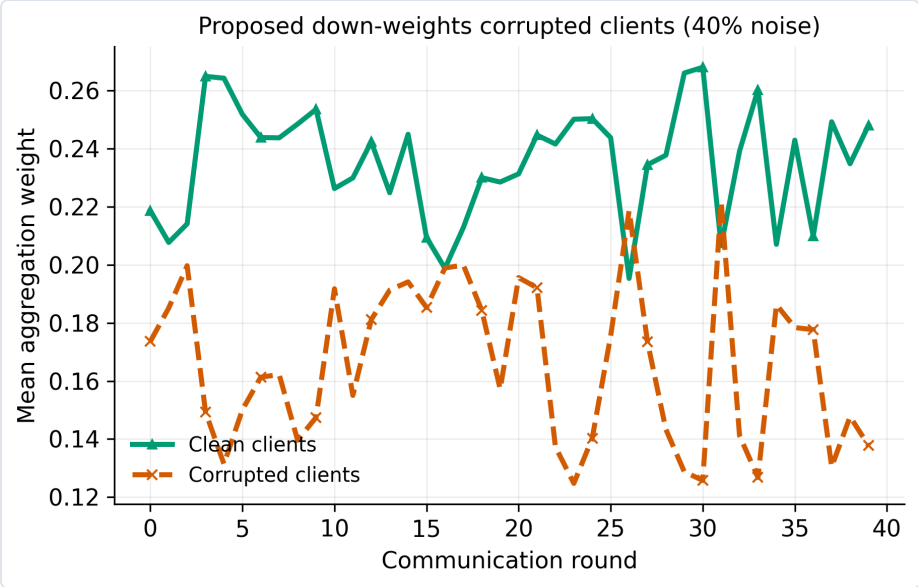


Figure 5. E3: mean aggregation weight assigned to clean vs corrupted clients across training. The clean line stays above the corrupted line throughout — corrupted clients are consistently down-weighted (ratio 0.69).

8.2 E4 — negative result: the ShapFed cosine signal fails

> Scoring clients by **cosine similarity** of the classifier update to the consensus (mean *or* robust coordinate-wise median) is **anti-discriminative** on OCTMNIST: corrupted clients get *higher* scores and *more* weight (noisy/clean weight ratio ≈ **1.1–1.3**, 3 seeds).
>
> **Cause.** Under partial participation (5 of 10 active, 40% corrupted), the active set often contains 2–3 corrupted clients, so the consensus direction is *itself* pulled toward the corrupted updates — the robust-median assumption breaks. **Cosine-of-update contribution scoring cannot identify label-noise clients under partial participation.** This motivated the loss detector.

8.3 E5 — stability: why moderate, not aggressive

Aggressive down-weighting ($\beta = 4$, cap = 0.5) collapses aggregation onto 1–2 clients and destabilizes the global model (best-round 0.57 but last-round crashes to 0.27–0.39 on some seeds). The moderate config (§4.2) removes the collapse (worst seed's last round 0.39 → 0.57). Hence the moderate configuration.

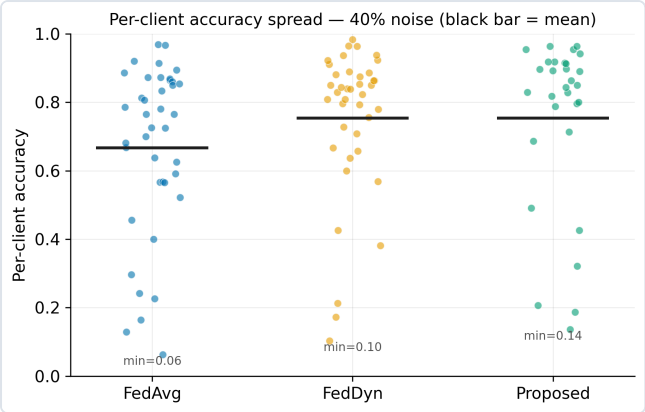


Figure 6a. Per-client accuracy spread at 40% noise — fairness view across all clients.

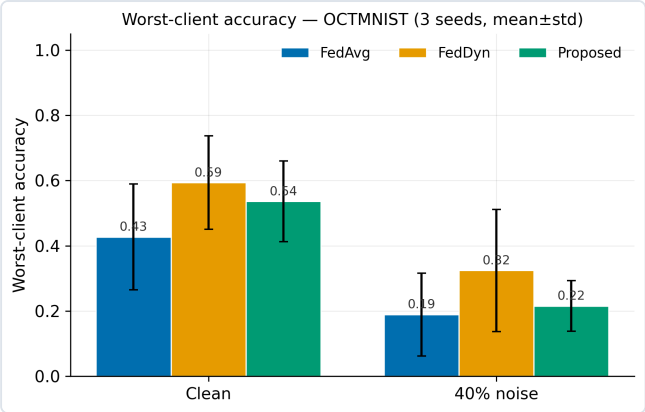


Figure 6b. Worst-client accuracy (OCTMNIST). FedDyn is fairer under noise.

9. The Central Finding — *When* Does It Help?

Down-weighting only pays off when the noise severely hurts the baseline. Define the **"opening"** = FedDyn's accuracy drop from clean → 40% noise:

Dataset	FedDyn opening (clean – 40%)	Proposed – FedDyn @ 40%
OCTMNIST	0.158 **(large)**	**−0.006 (near-tie)**
APTOS	0.066 (small)	**−0.061**
ISIC	0.062 (small)	**−0.027**

Interpretation. FedDyn's dynamic regularization already absorbs *moderate* label noise. When noise is mild (ISIC, APTOS) there is little to gain, and down-weighting throws away useful data → we lose. Only when noise is severe enough to break FedDyn (OCTMNIST) does down-weighting recover to parity — **but not beyond it**. Down-weighting's value is **conditional**.

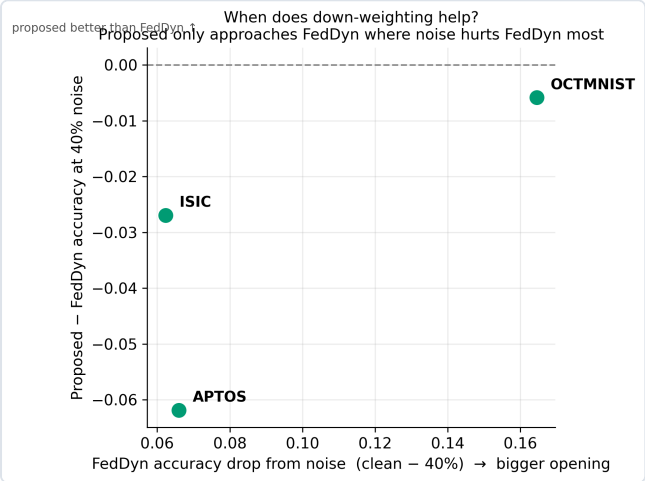


Figure 7a. Central finding: the Proposed–FedDyn gap approaches zero only as the 'opening' (FedDyn's noise-induced drop) grows.

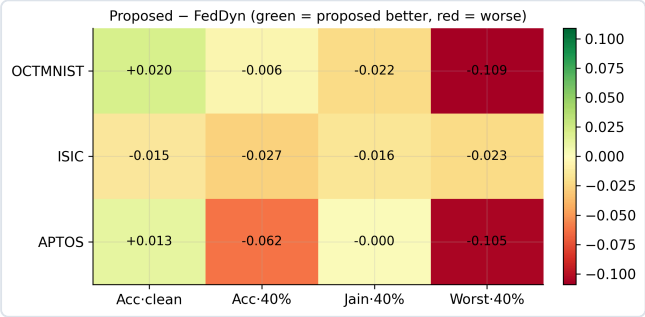


Figure 7b. Honest summary: Proposed–FedDyn across all metrics. Green (better) appears only on clean accuracy; every 40%-noise cell is ≤ 0.

10. Discussion, Honesty Statement & Reviewer Q&A

> **We DO claim:** (1) a loss-based client-quality detector with a *verified* mechanism (0.69 ratio); (2) a negative result — the ShapFed cosine signal fails under partial participation; (3) a multi-dataset characterization of *when* contribution-aware down-weighting helps.
>
> **We do NOT claim:** state-of-the-art accuracy, a general robustness win, or that the benefit generalizes. Making any of those claims would contradict our own tables (§7).

Anticipated reviewer questions

Question	Answer
Is the comparison fair?	Yes — one frozen protocol for all methods; the FedDyn baseline is provably unchanged by our weight-consistent integration; noisy-client sets are seeded identically.
Are the numbers cherry-picked?	No — we report the last (converged) round (cosine LR removes collapse), 3–5 seeds with std, and we show the *unfavorable* datasets, not only the favorable one.
Does the method do what it claims?	Yes — Fig. 5 directly verifies down-weighting of corrupted clients (0.69 ratio).
Then why isn't it a win?	FedDyn already tolerates moderate noise; down-weighting helps only when the baseline collapses, which is dataset-dependent (§9). This is a finding, not an experimental failure.
Is a negative / conditional result publishable?	Yes — a documented failure mode of a widely-used signal, a verified-but-conditional detector, and a reproducible characterization are useful knowledge for the community.

11. Limitations, Conclusion & Reproducibility

Limitations

- Two noise conditions (0%, 40%), label-noise only (no Byzantine, feature shift, or quantity skew).
- 3 seeds on ISIC/APTOS (5 on OCTMNIST clean); high variance under noise (APTOS proposed ±0.13).
- 64px inputs, 40-round budget (compute-limited, laptop MPS).
- Single participation regime (N=10, m=5).

Conclusion

A loss-based client-quality detector reliably identifies corrupted clients, and the resulting method is competitive with FedDyn on clean non-IID data. However, across three medical datasets, **down-weighting corrupted clients does not improve robustness over FedDyn under 40% label noise** — it only reaches parity where noise most degrades the baseline. We contribute this honest characterization, a documented failure of cosine-based contribution scoring, and fully reproducible code and logs.

Reproducibility

```
python3 scripts/run_noise_sweep.py --dataset octmnist --noises 0.0 0.4 --seeds
0 1 2
python3 scripts/run_noise_sweep.py --dataset isic     --noises 0.0 0.4 --seeds
0 1 2
python3 scripts/run_noise_sweep.py --dataset aptos    --noises 0.0 0.4 --seeds
0 1 2
python3 scripts/make_paper_figures.py        # OCTMNIST-detailed
python3 scripts/make_crossdataset_figures.py    # 3-dataset summary
```

Method preset: `config.py → METHOD_PRESETS["proposed"]` (FedDyn solver + `contrib_signal="loss"`). Ablation switches: `--contrib-signal cssv` (the failed signal), `--loss-weight-beta`, `--cssv-max-weight`, `--no-feddyn-weight-consistent`. Full tables with n per cell in `results/crossdataset_tables.md`.

Generated from the FedPowerShap repository (`docs/RESEARCH_STUDY.md`, `docs/SYSTEM_OVERVIEW.md`, `results/crossdataset_tables.md`, `figures/paper/`). All numbers trace directly to logged experiment runs.